between adjacent siblings, between parent and child elements with no padding, border, or content separating their margins, and in empty blocks. When margins collapse, the resulting space equals the largest margin (or most negative margin if negative), not their sum. This only affects vertical margins. For example, a 50px bottom margin and a 30px top margin between elements result in 50px of space, not 80px.

Margin collapsing can make spacing unpredictable, causing unexpected gaps or less space than intended. Adding padding, borders, or using overflow (not visible) prevents collapse. Margins do not collapse in flex or grid containers. Understanding margin collapsing is key for accurate layout design and troubleshooting spacing issues.

To implement a responsive layout where the box model behavior changes at different breakpoints, CSS media queries define key screen widths—such as 600px, 768px, or 1024px—where the layout or box model rules adapt. Within these media queries, properties like `box-sizing`, `padding`, or `margins` adjust to suit each breakpoint, and this is combined with modern layout techniques such as Flexbox or Grid for greater responsiveness. For example, setting `box-sizing: border-box` for mobile screens and switching to `box-sizing: content-box` for larger viewports ensures the layout remains consistent and adaptable across devices while maintaining control over how the box model affects element sizing and spacing.

For this topic, are you able to answer these follow-up/probing questions?

- "In what scenarios might you deliberately use `box-sizing: content-box` instead of `box-sizing: border-box`, and what are the trade-offs involved?"
- "How would you handle a situation where a legacy codebase uses `box-sizing: content-box` but you want to migrate to `box-sizing: border-box` for new components?"
- "What are the implications of using negative margins with the box model, and when might this technique be appropriate or risky?"
- "How do margins interact with the box model in the context of collapsing margins, and how does this affect layout calculations?"
- "How would you implement a responsive layout where the box model behavior needs to change at different breakpoints?"